S2S Lab 5

Exercise Solutions

Exercise 1

A university offers 8 different statistics courses to its honours level students. The numbers of students in each course are as follows:

- There are three course which have 85 students enrolled
- There are two course which have 112 students enrolled
- There is one course with 108 students enrolled
- There is one course with 95 students enrolled
- There is one course with only 46 students enrolled

Suppose you were to randomly sample three of these statistics courses **with** replacement. What would the sampling distribution of the sample mean number of students enrolled, $\bar{X}$, be? What would the sampling distribution of the sample variance in the number of students enrolled, S^2, be?

Can you use your sampling distributions to find $E[\bar{X}]$, $\sigma^2_{\bar{X}}$ and $E[S^2]$?

Solution

Create a vector `students` which contains the number of students enrolled in each of the 8 courses. This vector can be given as the arguments in `expand.grid()` to show all possible combinations of sampling three courses with replacement. There are 512 different combinations.

```
students <- c(rep(x = 85, times = 3), rep(x = 112, times = 2), 108, 95, 46)

samples_w <- expand.grid(Class1 = students, Class2 = students, Class3 = students)
```

Calculate the sample mean and sample variance for each possible sample of 3 courses. Update `samples_w` to contain these values and the sample they came from.

```
sample_mean_w <- apply(X = samples_w, MARGIN = 1, FUN = mean)

sample_var_w <- apply(X = samples_w, MARGIN = 1, FUN = var)

samples_w <- cbind(samples_w, Mean = sample_mean_w, Variance = sample_var_w)
```

Use `samples_w` to find the sampling distribution of $\bar{X}$.

```
samplingdist_mean_w <- table(samples_w$Mean)/512
```

Table 1: Sampling distribution for $\bar{X}$ when sampling with replacement.

$\bar{x}_i$	$P(\bar{X} = \bar{x}_i)$	$\bar{x}_i$	$P(\bar{X} = \bar{x}_i)$	$\bar{x}_i$	$P(\bar{X} = \bar{x}_i)$
46.000	0.0020	85.000	0.0527	100.333	0.0176
59.000	0.0176	87.333	0.0059	100.667	0.0117
62.333	0.0059	88.333	0.0527	101.667	0.0703
66.667	0.0059	88.667	0.0234	103	0.0703
68.000	0.0117	90.000	0.0234	103.667	0.0059
72.000	0.0527	91.667	0.0176	105	0.0234
75.333	0.0352	92.667	0.0527	106.333	0.0234
78.667	0.0059	94.000	0.1055	108	0.002
79.667	0.0352	95.000	0.0020	109.333	0.0117
81.000	0.0703	96.000	0.0352	110.667	0.0234
83.000	0.0117	97.333	0.0703	112	0.0156
84.333	0.0234	99.333	0.0059		

The expected value of $\bar{X}$ can be calculated as $E[\bar{X}] = 91$.

```
means_w <- sort(unique(samples_w$Mean))
mean_Xbar_w <- sum(means_w*samplingdist_mean_w)
mean_Xbar_w
```

```
[1] 91
```

The variance of $\bar{X}$ can be calculated as $\sigma^2_{\bar{X}} = 138.3333$.

```
var_Xbar_w <- sum(means_w^2*samplingdist_mean_w)-(mean_Xbar_w)^2
var_Xbar_w
```

```
[1] 138.3333
```

Use `samples_w` to find the sampling distribution of S^2.

```
samplingdist_var_w <- table(samples_w$Variance)/512
```

Table 2: Sampling distribution for S^2 when sampling with replacement.

s_i^2	$P(S^2 = s_i^2)$	s_i^2	$P(S^2 = s_i^2)$	s_i^2	$P(S^2 = s_i^2)$
0.000	0.0742	176.333	0.0703	982.333	0.0352
5.333	0.0352	186.333	0.0703	1069.000	0.0117
33.333	0.0703	212.333	0.0703	1101.000	0.0703
56.333	0.0117	243.000	0.1758	1174.333	0.0234
79.000	0.0234	507.000	0.0703	1281.333	0.0117
96.333	0.0352	670.333	0.0352	1369.333	0.0234
133.000	0.0352	800.333	0.0117	1452.000	0.0352

The expected value of the sample variance can then be calculated as $E[S^2] = 415$.

```
vars_w <- sort(unique(round(samples_w$Variance, 4)))
mean_S2_w <- sum(vars_w*samplingdist_var_w)
mean_S2_w
```

```
[1] 415
```

Exercise 2

Now suppose you want to randomly sample three of the above statistics courses **without** replacement. Find the sampling distribution of the sample mean number of students, $\bar{X}$ and the sampling distribution of the sample variance in the number of students enrolled, S^2.

Use these sampling distributions to find $E[\bar{X}]$, $\sigma^2_{\bar{X}}$ and $E[S^2]$.

How do these values compare to those found in Exercise 1?

Solution

Load the package `PASWR2` because sampling without replacement is used, so to list all the combinations of three class sizes that can be drawn, the `srs()` function is needed.

```
library(PASWR2)

samples_wo <- srs(popvalues = students, n = 3)
dimnames(samples_wo) <- list(NULL, c("Class1", "Class2", "Class3"))
```

Calculate the sample mean and sample variance for each possible sample of 3 courses. Update `samples_wo` to contain these values and the sample they came from.

```
sample_wo_means <- apply(X = samples_wo, MARGIN = 1, FUN = mean)

sample_wo_variances <- apply(X = samples_wo, MARGIN = 1, FUN = var)

samples_wo <- data.frame(samples_wo, Mean = sample_wo_means, Variance = sample_wo_variances)
```

Use `samples_wo` to find the sampling distribution of $\bar{X}$.

```
samplingdist_mean_wo <- table(samples_wo$Mean)/56
```

Table 3: Sampling distribution for $\bar{X}$ when sampling without replacement.

$\bar{x}_i$	$P(\bar{X} = \bar{x}_i)$	$\bar{x}_i$	$P(\bar{X} = \bar{x}_i)$
72.000	0.0536	92.667	0.0536
75.333	0.0536	94	0.1071
79.667	0.0536	96	0.0536
81.000	0.1071	97.333	0.1071
83.000	0.0179	101.667	0.1071
84.333	0.0357	103	0.0536
85.000	0.0179	105	0.0357
88.333	0.0536	106.333	0.0179
88.667	0.0357	110.667	0.0179
90.000	0.0179		

The expected value of $\bar{X}$ can be calculated as $E[\bar{X}] = 91$.

```
means_wo <- sort(unique(samples_wo$Mean))
mean_Xbar_wo <- sum(means_wo*samplingdist_mean_wo)
mean_Xbar_wo
```

```
[1] 91
```

The variance of $\bar{X}$ can be calculated as $\sigma^2_{\bar{X}} = 98.8095$.

```r
var_Xbar_wo <- sum(means_wo^2*samplingdist_mean_wo)-(mean_Xbar_wo)^2
var_Xbar_wo
```

```
[1] 98.80952
```

Use `samples_wo` to find the sampling distribution of S^2.

```r
samplingdist_var_wo <- table(samples_wo$Variance)/56
```

Table 4: Sampling distribution for S^2 when sampling without replacement.

s_i^2	$P(S^2 = s_i^2)$	s_i^2	$P(S^2 = s_i^2)$
0.000	0.0179	243.000	0.1607
5.333	0.0179	507.000	0.0536
33.333	0.0536	670.333	0.0536
79.000	0.0357	982.333	0.0536
96.333	0.0179	1069.000	0.0179
133.000	0.0536	1101.000	0.1071
176.333	0.0536	1174.333	0.0357
186.333	0.1071	1369.333	0.0357
212.333	0.1071	1452.000	0.0179

The expected value of the sample variance can then be calculated as $E[S^2] = 474.2857$.

```r
vars_wo <- sort(unique(round(samples_wo$Variance, 4)))
mean_S2_wo <- sum(vars_wo*samplingdist_var_wo)
mean_S2_wo
```

```
[1] 474.2857
```